Experiment - 7

DAA

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- Algorithm

START

Step-I: Initialise costs,

dist[src] = 0 and all other series to ∞ (or INT_MAX)

Step-II: Relax edges upto k+1 times (since k stops means k+1 edges)

i) For each iteration, copy current distances into a temporary array.

ii) For each flight, update:

temp[v] = min(temp[v], dist[u] + price)

iii) After processing all flight, set dist = temp

Step-III: After k+1 iterations, dist[dst] gives cheapest cost.

i) If still INT_MAX or ∞, return -1.

- Dry Run.

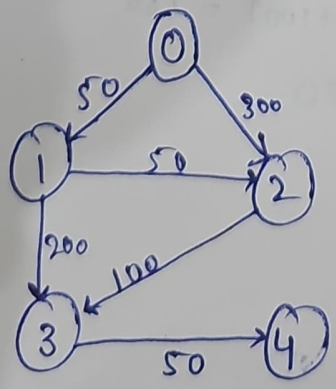
n = 5

flights = [[0,1,50], [0,2,300], [1,2,50], [1,3,200], [2,3,100], [3,4,50]]

src = 0

dst = 4

k = 2



Step-0

dist = [0, ∞, ∞, ∞, ∞]

Step-1

0 → 1, 50

0 → 2, 300

update these in dist.

dist = [0, 50, 300, ∞, ∞]

Step-2

0 → 1, 50

0 → 2, 300

1 → 2, 50 [min(300, 50+50)] = 100 ← dist[2]

1 → 3, 200 [temp[3] = 50+200 = 250

2 → 3, 100 [temp[3] = min(250, 300+100) = 250]

3 → 4, 50 ignored.

Step-3

0 → 1, 50 temp[1] = 50

0 → 2, 300 temp[2] = 100

1 → 2, 50 temp[2] = 100

1 → 3, 200 temp[3] = 250

2 → 3, 100 temp[3] = min (250, 100+100) = 200

3 → 4, 50 temp[4] = 200 + 50 = 250

dist = [0, 50, 100, 200, 250].

Cheapest cost = 250.